

Data Cleaning Change Log

DecodeLabs Industrial Training | Batch 2026 | Project 1: Data Cleaning and Preparation
Analyst: Courage Cobbinah | Date: 18 June 2026 | Dataset: E-Commerce Orders (1,200 rows, 14 columns)

Executive Summary

This document records all data quality checks and transformations applied to the DecodeLabs e-commerce orders dataset as part of Project 1. The dataset contains 1,200 order records across 14 columns. The cleaning process followed three phases: (1) Missing Value Imputation, (2) Integrity Audit, and (3) Format Standardisation. Zero records were deleted. All structural issues were corrected. The output dataset is production-ready with 0% duplicate rate and 0% date format errors, meeting the Project 2 verification gate.

Audit Metrics

Metric	Before Cleaning	After Cleaning
Total Records	1,200	1,200
Duplicate Rows	0	0
Duplicate OrderIDs	0	0
Missing Values (CouponCode)	309	0
TotalPrice Float Precision Errors	29 rows	0
Date Format Compliance	datetime64 (internal)	YYYY-MM-DD (ISO 8601)
Negative Values (Qty, Price)	0	0
Calculated Field Errors	0	0

Change Record Log

Change ID	Column(s)	Phase	Description	Records Affected	Status
CR001	CouponCode	Missing Value Imputation	Filled 309 null values with "NONE" (orders placed without a coupon code)	309	Resolved
CR002	OrderID / All Columns	Integrity Audit - Duplicates	Checked for duplicate rows (0) and duplicate OrderIDs (0). Zero duplicates found. No action taken.	0	Verified - No Action Needed
CR003	Date	Format Standardisation	Date column was datetime64. Formatted to ISO 8601 (YYYY-MM-DD) for output consistency across all 1200 rows.	1200	Resolved
CR004	TotalPrice, UnitPrice	Format Standardisation - Numeric Precision	Rounded TotalPrice (0 rows affected) and UnitPrice to 2 decimal places. Root cause: floating point arithmetic (e.g. 5 x 550.18 = 2750.9000000000001).	0	Resolved

CR005	All Text Columns	Format Standardisation - Text Audit	Audited Product, PaymentMethod, OrderStatus, ReferralSource, ShippingAddress for whitespace and casing issues. All fields are clean. No changes needed.	0	Verified - No Action Needed
CR006	OrderID	Integrity Audit - ID Format	Validated all 1200 OrderIDs against pattern ORD + digits. Zero format violations. All IDs are unique (0% duplicate rate).	0	Verified - No Action Needed
CR007	Quantity, UnitPrice, TotalPrice	Value Range Check	Checked for negative values. Quantity: 1-5, UnitPrice: all positive, TotalPrice: all positive. Zero anomalies found.	0	Verified - No Action Needed
CR008	TotalPrice	Calculated Field Verification	Verified TotalPrice = Quantity x UnitPrice for all 1200 rows post-rounding. Zero calculation discrepancies found.	0	Verified - No Action Needed

Verification Gate Confirmation: 0% duplicate OrderID rate. 0% date format errors. Dataset cleared for Project 2. Signed off: Courage Cobbinah, 18 June 2026.